The Price of Early Data:

How State GDP Revisions Would Have Shifted ₹25,654 Crore Between India's States

Evidence from ITIHAS, a Real-Time Database of Indian State GDP

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Abstract

India uses state GDP to calculate the split of central tax money between states and each state's borrowing cap. Both of these are set using GDP figures that keep changing for years after they're first published. This led us to ask how much money has moved between states because the tax split and borrowing limits use GDP figures that have not settled, and what change to the rules would reduce the movement. We built an Indian Time-series of Initial and Historical Accounts of States (ITIHAS), which records published estimates of each state's GDP with the date it came out: 3,631 figures, 431 published versions, 20 states, 2012-13 to 2025-26. We found the two-year rule where early figures get states' relative growth wrong for about two years; the correlation between early and final figures is 0.22–0.40 before two years and 0.89 after. The problem cannot be solved by predicting the final numbers early as none of our 19 models designed in advance beat the national rate. The study found that ₹25,654 crore of the 15th Commission's 2021-26 award would have gone to different states, and that about ₹3,600–5,000 crore a year is normal across past Commissions. Borrowing caps were ₹1.44 lakh crore higher than corrected figures give. We suggest that the fix is not using older data, as the settled data worsens the error, but rather re-running the formula with the corrected figures in year 3 of the award (a "true-up"), which fixes about 44% of the moving money. The same idea works for the borrowing caps.

Keywords: real-time data; data revisions; state GDP (GSDP); fiscal transfers; Finance Commission; fiscal federalism; nowcasting; official statistics; borrowing limits; India.

JEL codes: H77, E01, C82, C53, R11, O53.

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1. Introduction

The 15th Finance Commission set how central tax money in India is split among its states for FY 2021-22 to 2025-26. If its formula had used the corrected state GDP (gross state domestic product, GSDP) figures instead of the early ones, ₹25,654 crore over those five years would have gone to different states. Uttar Pradesh and Bihar would have got ₹9,617 crore and ₹8,467 crore more respectively, whereas Tamil Nadu would have received ₹6,228 crore less. The formula depends on each state's GDP. The Commission reported in November 2020 using GDP for 2016-17 to 2018-19, when the 2018-19 figures were under two years old. States kept correcting them after that. States together receive 41% of shared central taxes: ₹15.3 lakh crore planned for 2026-27 (Ministry of Finance, 2026a; 2026b). The biggest part of the split goes to states whose GDP per person is furthest below the richest states: 45% of the total under the 15th Commission, 42.5% under the 16th. Each state's yearly borrowing limit is also 3% of its projected GDP, about ₹10 lakh crore for all states together in 2024-25 by our calculation. India splits tax money and sets borrowing limits using state GDP figures that are still being corrected. This study therefore asks how much money has moved between states because the tax split and borrowing limits use GDP figures that have not settled, and what change to the rules would reduce the movement.

State GDP figures are repeatedly corrected for two to three years after the year ends, with early figures missing turns in the economy. For 2019-20, which was a slowdown, they were too high and were cut by 2.25 points on average, with 93% of states revised down. For 2023-24, which was a pickup period, they were too low and were raised by 0.93 points. The corrections are bigger than the gaps between states; a typical correction is about 2.7 points, while states' first growth figures differ from each other by about 1.8 points. A correction can easily change which states look the fastest. Yet the money is decided on the early figures and then locked in. The Commission's split stays fixed for five years, and borrowing limits are never recalculated even when the figures are corrected. Frankel (2011) and Frankel and Schreger (2013) show that governments' growth forecasts tend to be too optimistic, which helps them get around deficit limits. The problem in this case is different, as Indian state budgets' GDP assumptions are not too high on average (Table A4); it is the "actual" past figures used by the rules that keep changing for about two years. Australia's Grants Commission reports how much corrected data moves money between its states every year, and Mitchell and Shiroff (2025) study corrections to US state GDP. To our knowledge, no one has done this for India, as nobody kept a record of which figure was published when.

We built an Indian Time-series of Initial and Historical Accounts of States (ITIHAS), which records published estimates of each state's GDP with the date it came out: 3,631 figures, 431 published versions, 20 states, 2012-13 to 2025-26. It is built upon 215 documents from 18 states' own statistical offices (some were scans read with text-recognition software), 9 editions of RBI's Handbook of Statistics on Indian States, and 2 releases from MoSPI (the national statistics ministry). Each number links to the page it came from, and the database is free on GitHub. Every analysis rule was written down before we saw any results, and all the changes were logged. A sample of the scanned numbers was checked by hand against the original pages.

Our first finding is "the two-year rule". For about two years after a year ends, early figures rank states' growth badly. For a state's growth compared with the all-state average, the correlation between early and final figures is 0.22–0.40 before two years and 0.89 after (Figure 3). The jump is statistically clear: +0.49, with a 95% range of 0.16 to 0.83 (one-sided $p<0.001$). When numbers are first published, 37% of figures put a state on the wrong side of the all-state average, showing that it grew faster than average when it actually grew more slowly, or the reverse (Figure 4).

We tried 19 ways of nowcasting the final GDP figures, each designed before seeing the actual results. The models used GST collections, e-way bills, electricity use, satellite night lights, mining output and states' own early estimates. None beat simply assuming every state grows at the national rate, and for the best models any improvement bigger than about 11–18% is ruled out (Table 5).

We found that about 78% of the ₹25,654 crore comes from the newest year the Commission used (2018-19), which is the youngest figure (Table 6). We re-ran the formula as if past Commissions had used older editions of the RBI data, which we call "placebo Commissions". The normal amount that moves is about ₹3,600–5,000 crore a year. For the current Commission (the 16th, 2026-31), we expect ₹3,400–3,900 crore a year to move, and ₹1,110 crore a year has already moved after only 8 months of corrections. India is also about to change the base year it uses to measure GDP, and if states shift as much as in past changes, it could cause ₹13,000–22,000 crore a year to move. No state can tell in advance whether it will gain or lose (Table 8). Borrowing limits built on the early GDP figures added up to ₹1.44 lakh crore more over 2019-20 to 2024-25 than the corrected figures would give, mostly in 2020-21 to 2022-23. A typical state's limit changes by 4.6% in a year depending on which version of GDP is used (Table 9).

As none of our 19 models could predict the final figures early, the fix must come from when the rules use the data. The obvious fix, which is using older and more settled years, does not work. We tested it with a rule set in advance. Using data one year older gives a higher error (₹6,315 against ₹5,687 crore a year) and is better in only 1 of 5 past cases, because states' relative incomes change about as fast as the figures get corrected (Figure 8). The better fix is what we call a "true-up". Partway through the five years, the Commission re-runs the same formula with the corrected figures for the same years, and uses the new shares for the remaining years. Done in year 3, this fixes about 44% of the money that would otherwise move (Table 10). The same idea works for borrowing: recalculate the limits once the figures are corrected. Before the new base-year GDP is used for any money decision, publish a table linking old and new figures and calculate shares both ways.

The closest Indian study is Ghosh, Singh and Kuruganti (RBI Bulletin, October 2026). They estimate current activity for 3 states using night lights and Google searches, on the latest data. We test against what was actually known at the time, for 20 states, and add the money question. Narayana (2024) argues the formula is sensitive to base-year changes but does not recalculate shares. ICRA (2022) warns that corrections can push states over borrowing limits but does not recalculate limits state by state. To our knowledge, ours is the first database of Indian state GDP that records when each figure was published, and the first measure of how much money moves in India's tax split and borrowing limits because

of corrections. The paper will now provide background, data, approach, the two-year rule, nowcasts, the money, the fix, the limitations, and conclusion.

2. Background

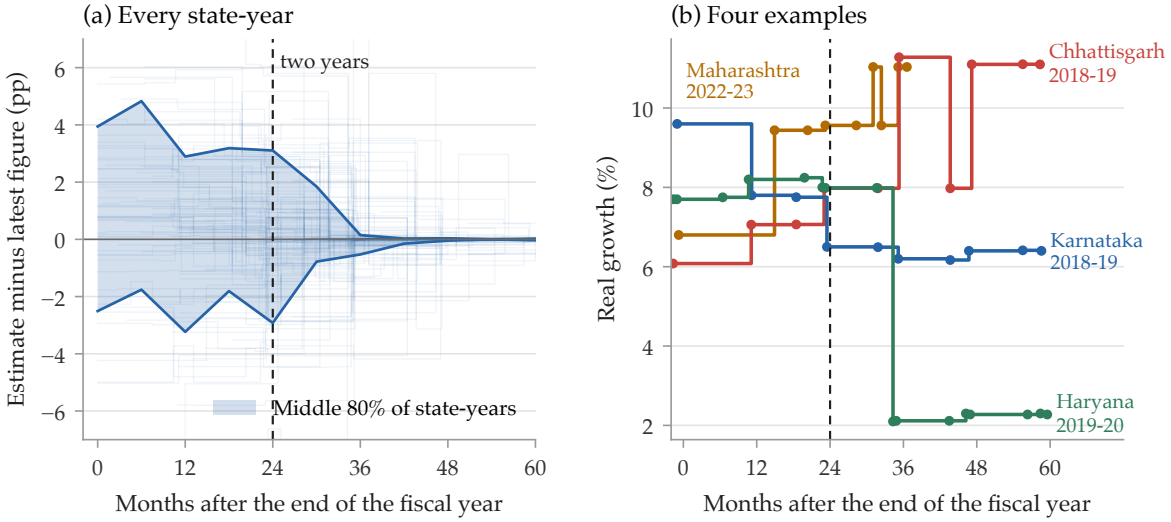
2.1 How state GDP is compiled and published

The state-wise GDP is calculated by each state's own statistics office (DES, Directorate of Economics and Statistics), which uses the national methodology of the National Statistics Office (NSO), part of the Ministry of Statistics and Programme Implementation (MoSPI). In principle, the estimates are therefore comparable across states. The first figure, or the "advance estimate", is published around February–March, before the financial year ends on 31 March, and is built from early indicators such as crop forecasts, budget spending, and tax receipts. Where such indicators are missing, some states, such as Telangana, use the all-India growth rate (Directorate of Economics and Statistics, Telangana, 2019). This is followed by the quick, provisional, and revised figures over the course of 2–3 years, with each new number replacing the previous. Not every state follows this sequence, such as Kerala, which has no advance estimate at all. Figure 1 shows how far a figure can move once it is published. Karnataka's growth for 2018-19 was first published at 9.6% and now stands at 6.4%, whereas Chhattisgarh's rose from 6.1% to 11.1%. MoSPI collects all states into one table twice a year in March and August (Fourteenth Finance Commission, 2015, para 14.65), with the RBI Handbook reprinting a MoSPI snapshot. Since the Handbook, one of our three sources, reprints an older table (its December 2025 edition carries MoSPI's August 2025 table), it can lag months behind the states' own releases. The largest revision is the base-year change, where every price and weight is recalculated from a new reference year. The previous one was from 2004-05 to 2011-12, and the next one is the 2022-23 base. The national series was released on 27 February 2026 (Ministry of Statistics and Programme Implementation, 2026a), followed by the state guidelines on 7 May 2026 (Ministry of Statistics and Programme Implementation, 2026c). The state numbers are not yet out, leaving a major revision still to come, as every state's GDP history will be revised again (Section 7.3).

2.2 Where state GDP enters fiscal rules

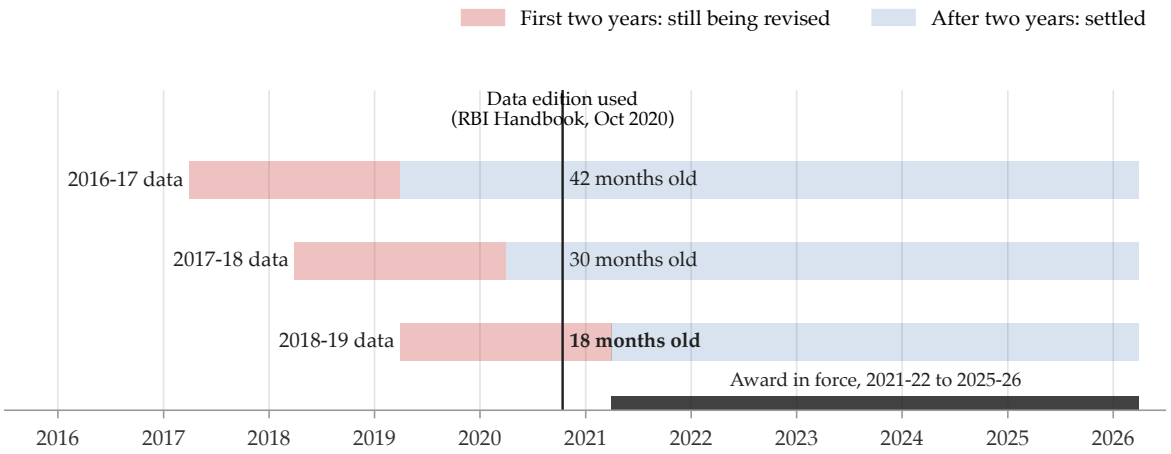
The reason the money moves is that the Centre collects more tax than it spends whereas states spend more than they collect. The Finance Commission is a constitutional panel set up every five years, which decides the states' total share and how the money is split between them. The splits depend on a formula that encompasses multiple factors such as population, area, forest cover, and others (Fifteenth Finance Commission, 2020; Sixteenth Finance Commission, 2025). The most relevant factor is income distance, where states with a lower GDP per capita get a larger proportion. The GDP figures are taken only once when the report is written, and then used for the entire five-year period. Figure 2 shows how old the 15th Commission's data were. The Commission used the RBI Handbook of October 2020, when its newest year, 2018-19, was only 18 months old and therefore still being corrected. The only changes that matter are relative changes, because if every state is revised by the same %, then nothing moves between them. This is the reason why the paper looks at each state compared to the average.

Figure 1. How far published state growth figures move after release



Notes: Panel (a) has one line per state-year (173 state-years, 20 states, FY2012-13 to 2019-20 and 2022-23, COVID years excluded), showing each published estimate minus the latest figure. The shaded band is the middle 80% in six-month steps. Panel (b) shows four state-years, where each dot is one dated release. The dashed line marks two years after the end of the fiscal year. Source: ITIHAS; authors' calculations.

Figure 2. Age of the GSDP data the 15th Finance Commission used



Notes: Age is counted from the end of each fiscal year to 13 October 2020, the date of the RBI Handbook edition whose GSDP the Commission used. The two-year line comes from Section 5. The award applies to 2021-22 to 2025-26. Source: Fifteenth Finance Commission (2020); Reserve Bank of India (2017–2025); authors' calculations.

The borrowing cap is a yearly limit on how much a state can borrow as a % of its GDP. This is usually set at 3%, but was changed to 4% in 2021-22 and 3.5% in 2022-23, not counting 2020-21's extra 2% for COVID or the 0.5% power-sector allowance. The cap uses projected GDP as the current year's GDP does not yet exist when the cap is set. The projection takes the GDP from two years earlier and increases it for two years at the state's recent average growth (Fourteenth Finance Commission, 2015). The cap is also kept constant even when the GDP figure is revised.

3. Data

ITIHAS is a record of 431 published versions of each state's GDP, saved with the date each first became public, covering 20 states from 2012-13 to 2025-26 (Table 1). It has both current-price and constant-price figures divided roughly equally. Data has been obtained from 215 documents from 18 states' statistics offices, economic surveys, and budget papers. In totality, that is 1,593 figures of which 1,344 were read as text and 249 came from scans read by text-recognition software. Furthermore, 9 editions of the RBI Handbook of Statistics on Indian States and 2 MoSPI releases from August 2025 and April 2026 were also used to compile the data (Reserve Bank of India, 2017–2025; Ministry of Statistics and Programme Implementation, 2025; 2026b). Each figure links to its document and page number and is dated to the day it first became public, which is the date printed on the document. For budget papers, the date is the day the budget was presented. When two releases carry the same date and disagree, we keep the one from the state's statistics office or Economic Survey over budget documents and press notes.

Table 1. Coverage of ITIHAS by state

State	Own releases	Own releases, first – last	Versions (all sources)	Values in panel	of which from own releases	of which read by OCR
Andhra Pradesh	12	Mar 2017 – Jun 2026	23	169	63	0
Assam	10	Aug 2017 – Feb 2026	20	159	73	37
Bihar	9	Feb 2019 – Apr 2026	20	198	93	53
Chhattisgarh	18	Mar 2017 – Feb 2026	29	171	66	53
Gujarat	17	Feb 2018 – Feb 2026	28	221	125	9
Haryana	17	Mar 2017 – Jan 2026	27	260	169	14
Himachal Pradesh	4	Mar 2023 – Mar 2026	15	153	50	0
Jharkhand	7	Aug 2021 – May 2025	18	170	66	12
Karnataka	10	Feb 2018 – Mar 2026	21	210	104	0
Kerala	20	Jan 2019 – Jan 2026	31	229	129	45
Madhya Pradesh	0	none	11	106	0	0
Maharashtra	10	Mar 2018 – Mar 2026	21	177	77	0
Odisha	16	Mar 2018 – Mar 2026	27	264	157	8
Punjab	10	Mar 2018 – Mar 2026	21	198	96	7
Rajasthan	10	Mar 2017 – Feb 2026	21	200	95	8
Tamil Nadu	13	Mar 2017 – Feb 2026	24	138	32	3
Telangana	12	Mar 2017 – Mar 2026	23	188	82	0
Uttar Pradesh	3	Feb 2019 – Feb 2026	14	122	17	0
Uttarakhand	17	Jul 2016 – Feb 2026	28	203	99	0
West Bengal	0	none	9	95	0	0
All 20 states	215	Jul 2016 – Jun 2026	431	3,631	1,593	249

Notes: Real-time panel of constant-price GSDP growth, FY2012-13 to 2025-26. A version is one dated publication. Each version counts once per state. Source: ITIHAS (state statistics releases, 9 RBI Handbook editions, 2 MoSPI releases); authors' calculations.

Given that West Bengal and Madhya Pradesh have no state release, we use only the central versions. Therefore, for a few states the early history comes only from central sources.

First, we verified the printed growth rates against growth calculated from printed levels, obtaining 99.8% of 2,223 pairs agreeing within 0.06 points (Table 2). Secondly, we checked 31

scanned values against the original pages by hand. Thirty of the figures matched and the 31st was a blank cell (Kerala, 2017-18), which was filled from the page (Table 2). We measured each state's growth compared with the all-state average, because Section 2.2 showed only relative changes move money. Age is the number of months between the end of the financial year and the date the figure was published, and the final figure is the most recent one present in the database. The COVID years, 2020-21 and 2021-22, are left out of the timing analysis due to the pandemic.

Table 2. Data checks

Check	Count	Result
Rows from states' own releases (18 states), both price bases	3,585	
read from the PDF text layer	3,028	every value found verbatim in its source
read by OCR or by eye from scanned pages	557	each carries a page marker
Printed growth vs growth computed from printed levels (same table)	2,223	99.8% within 0.06 pp; 100.0% within 0.5 pp
Hand check against the source page	31	30 matched; 1 omitted value found and corrected
Rows with a printed growth rate but no level	36	documents that print only a growth rate (budget speeches, press notes)
OCR blanks found and filled from the page (9 Nov)	12	Kerala, three releases; each reproduces the printed growth
State-years printed by two releases on the same date	43	4 differ: 1 substantive (Chhattisgarh 2019-20), 3 from rounding
Central versions matched to a state release	100 of 196	median lag 226 days (IQR 98–278); a newer state figure was skipped in 16%
Gap between a central figure and the state's newest figure at that date	146	mean absolute gap 0.42 pp; 12% exceed 1 pp

Notes: Counts refer to the full database (both price bases) unless the row says otherwise. Source: ITIHAS; authors' calculations.

4. Approach

When the Commission chooses which years of GDP to use, it faces two types of errors. Revision error comes from figures that are still being corrected, and staleness comes from older figures that no longer match states' current relative incomes. Revision error reduces as age increases, and falls sharply after two years. Revisions to a state's relative income have a spread of about 4.7–4.9% when the data are under two years old, 2.9% at two to three years, and 0.6% after three years. Staleness is a state's relative income moving by about 4.3% in a typical year. Using data one year older swaps one error for another of about the same size, therefore providing no gain. This can be studied through the following equation:

$$V(\ell) = R(\ell) + S(\ell), \qquad S(\ell) = \mathrm{Var}_s\left(x_{s,t} - x_{s,t-\ell}\right) \tag{1}$$

$V(\ell)$ is the total error when the data used are ℓ years older than the Commission's, which is the thing being explained. $R(\ell)$ is the revision error and $S(\ell)$ is the staleness, which is how

much a state's relative income $x_{s,t}$, the log of its GDP per person minus the all-state average, changes over ℓ years. Using older data only helps if the revision error it removes, $\mathrm{R}(0) - \mathrm{R}(\ell)$, is bigger than the staleness it adds, $\mathrm{S}(\ell)$. One year older removes about 0.0003–0.0006 of revision variance but adds about 0.0006 of staleness, which is why it does not help. Re-measuring the same years later (a true-up) cuts revision error without adding staleness. Our simple model of the two errors ranks the Commission's current window first and puts the best true-up in year 3. The direct money tests in the fix section agree (Table 10).

The two-year rule is measured by the correlation between early and final relative growth for each age group. This can be studied through the following equations:

$$d^{v}_{st} = g^{v}_{st} - \bar{g}^{v}_{t} \tag{2}$$

$$\rho(A) = \mathrm{Corr}\big(d^{v}_{st},\, d^{L}_{st}\big) \text{ for } a_{v,t} \in A, \qquad \Delta = \rho_{2\text{–}3\,\mathrm{yr}} - \rho_{1\text{–}2\,\mathrm{yr}} \tag{3}$$

d^{v}_{st} is the growth of state s in year t as printed in release v, minus the average growth across all states for that year, $\bar{g}^{v}_{t}$. $\rho(\mathrm{A})$ is the number the study cares about, which is the correlation between this early relative growth and the final one, d^{L}_{st}, for every release whose age $a_{v,t}$ falls in age group A. Δ is the jump between the one-to-two-year group and the two-to-three-year group. The uncertainty comes from a bootstrap that redraws whole states 2,000 times, so each state's revisions stay together. The p-value is the share of the 2,000 draws in which Δ is zero or below. Prior to changing anything, we rebuilt each Commission's formula from its published inputs and reproduced its published shares to within 0.001 percentage points, which shows our formula is the real one. We scale each GDP figure the Commission used by its later revision, from the RBI edition it used to the latest figure, and keep every other input the same and re-run the formula. This can be studied through the following equations:

$$\tilde{G}_{sy} = G^{FC}_{sy} \times \frac{L_{sy}}{V_{sy}}, \qquad \bar{y}_{s} = \frac{1}{3}\sum_{y} \frac{\tilde{G}_{sy}}{N_{sy}}, \qquad D_{s} = (\bar{y}_{b} - \bar{y}_{s})\, P_{s} \tag{4}$$

$$\sigma_{s} = \sum_{k} w_{k}\, c^{k}_{s}, \qquad c^{ID}_{s} = \frac{D_{s}}{\sum_{j} D_{j}} \tag{5}$$

G^{FC}_{sy} is the GDP of states in year y as the Commission used it, and $\tilde{G}_{sy}$ is the corrected version. V_{sy} is the same figure in the October 2020 RBI Handbook and L_{sy} is the latest figure. $\bar{y}_{s}$ is the three-year average GDP per person, N_{sy} is population, b is the benchmark state, which is Haryana for the 15th Commission, and P_{s} is the 2011 population. σ_{s} is a state's share of the money, which adds up its shares c^{k}_{s} on every criterion k with weights w_{k}. As each state's income distance is divided by the sum of all states' distances, a revision to one state's GDP changes every state's share. Money that moved is half of the total change in states' shares times the money actually paid out.

$$M = \frac{1}{2}\sum_{s} |\tilde{\sigma}_{s} - \sigma_{s}| \times T \tag{6}$$

where T is the money actually transferred over 2021-26. The reason we take half is that one state's gain is another state's loss, so this is done for the purpose of not counting twice. To

answer the question whether the final figures could be predicted, we ran nowcasts with different models. Models only use what was public by the end of February of that fiscal year, a month before it ends (models using a state's own advance estimate may use one published up to 30 April after the year), which is when the Commission would need the number. The benchmark is every state growing at the national first estimate, which is the simple guess every model has to beat. The national First Advance Estimate, released each January, is the newest national figure available at that date. Each of the 19 models was written down before running. These were built on states' early estimates, GST, e-way bills, electricity, night lights patterns, and mining, among others. We scored our prediction against actual final figures from 2018-19 to 2024-25, excluding the COVID years. We tested them through squared errors compared to the benchmark (Diebold and Mariano, 1995). This can be studied through the following equation:

$$e_{m,st} = \hat{g}_{m,st} - g^L_{st}, \qquad e^2_{m,st} - e^2_{0,st} = \mu_m + u_{m,st} \tag{7}$$

$e_{m,st}$ is the error of model m for state s in year t against the final figure, and model 0 is the benchmark. μ_m is the number the study cares about, which is how much more squared error model m makes than the benchmark on average, such that a negative value means the model is more accurate. $u_{m,st}$ is everything else the model does not capture. Standard errors are clustered at the state level, as the same state's errors are related across years. A "win" or more accurate prediction is one which is still statistically significant after the correction for trying 19 models (Holm, 1979).

5. The two-year rule

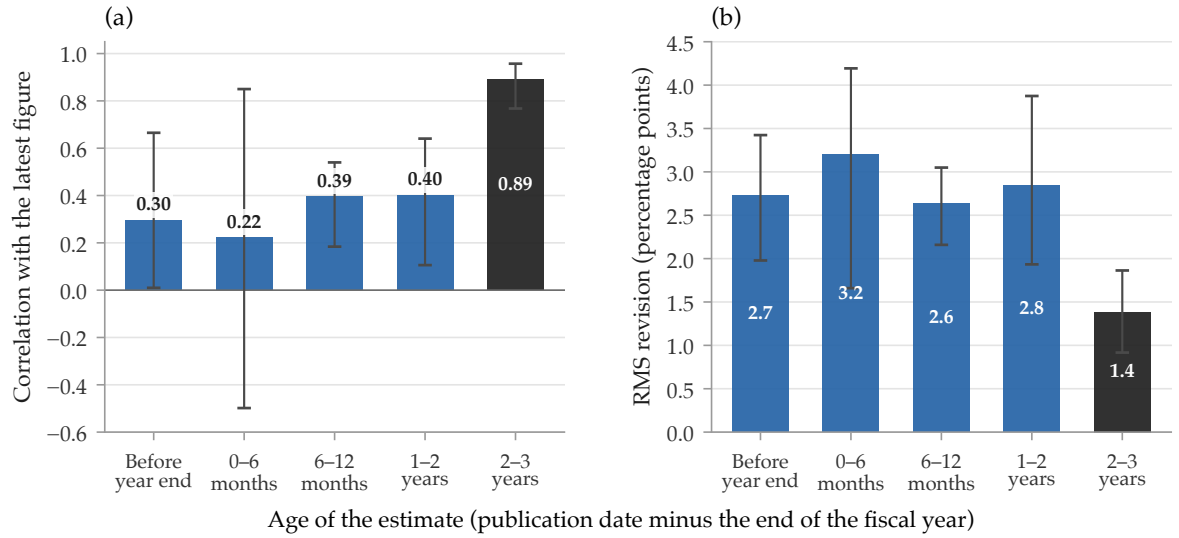
This section answers how old a figure has to be before it compares states correctly. The correlation stays between 0.22 and 0.40 for the first two years, and jumps to 0.89 after (Figure 3). The jump is +0.49, with a 95% interval of 0.16 to 0.83 (one-sided $p < 0.001$) (Table 3). This means that the jump is not gradual, as it is flat for two years and then one large step, which matches Section 2.1. The revised estimates, which are built on fuller data, arrive about two years after the year ends.

Table 3, Panel B is the stability check, where once figures are at least 3 years old, every RBI edition agrees with the latest one (0.94–1.00). Therefore the final figure we compare against is itself stable.

The typical correction is 2.70 points, which is bigger than the 1.78-point spread between states in the first figures. Thirty-seven percent of first figures are on the wrong side of the average (Figure 4). Of the fastest five states in the first figures, only 2 to 4 stay in the top five once the figures are revised. The direction of the errors changes each period, where although 2019-20 was a slowdown with early figures cut by 2.25 points and 93% of the states revised down, 2023-24 was a pickup with early figures raised by 0.93 points (Table 4). Whether revisions are new information or corrections of mistakes can be tested through the following equation (Mankiw and Shapiro, 1986):

$$r_{st} = g^L_{st} - g^F_{st}; \qquad \textit{news: } \mathrm{Corr}(r, g^F) = 0, \qquad \textit{noise: } \mathrm{Corr}(r, g^L) = 0 \tag{8}$$

Figure 3. How well early figures match the latest figure, by age of the estimate



Notes: Bars: estimate. Whiskers: 95% interval from a state-cluster bootstrap (2,000 draws). 20 states, FY2012-13 to 2019-20 and 2022-23 (COVID years excluded). Source: ITIHAS; authors' calculations.

Table 3. Accuracy of early figures by age of the estimate

Age of the estimate	n	Correlation with latest	95% interval	RMS revision (pp)	95% interval / note
Panel A. Correlation and revision size by age of the estimate					
Before year end	43	0.30	[0.01, 0.67]	2.73	[1.98, 3.42]
0–6 months	19	0.22	[−0.50, 0.85]	3.20	[1.66, 4.19]
6–12 months	86	0.39	[0.18, 0.54]	2.64	[2.16, 3.05]
1–2 years	111	0.40	[0.11, 0.64]	2.84	[1.93, 3.87]
2–3 years	129	0.89	[0.77, 0.96]	1.38	[0.92, 1.86]
Jump, 2–3 years minus 1–2 years		0.49	[0.16, 0.83]		one-sided $p < 0.001$
Panel B. Test–retest, main 20 states: each edition vs the Dec 2025 edition (state-years at least 3 years old)					
Handbook Jun 2017	34	0.988		0.36	2.6% of variance
Handbook May 2018	51	0.938		0.83	12.2% of variance
Handbook Mar 2019	54	0.998		0.13	0.3% of variance
Handbook Oct 2020	94	0.988		0.46	2.4% of variance
Handbook Nov 2021	94	1.000		0.06	0.0% of variance
Handbook Nov 2022	94	0.998		0.16	0.3% of variance
Handbook Nov 2023	154	0.998		0.18	0.5% of variance
Handbook Dec 2024	154	1.000		0.07	0.1% of variance

Notes: Panel A: correlation of cross-state growth deviations with the latest figure; RMS of the revision to growth. Intervals: state-cluster bootstrap, 2,000 draws. Panel B: correlation of cross-state growth deviations between an earlier edition and the Dec 2025 edition; SD of the late revision (pp) and its share of cross-state variance. All states and UTs: 0.97–1.00. Source: ITIHAS; authors' calculations.

r_{st} is the revision, which is the final growth minus the first. If revisions are new information, they cannot be predicted from the first figure, so their correlation with it is zero. If revisions only fix mistakes, their correlation with the final figure is zero. The revisions match the final figure (0.79), not the first one (−0.13). Therefore, they are new information rather than mistakes being fixed (Mankiw and Shapiro, 1986). Therefore, early figures cannot be statistically cleaned, and the 15th Commission's newest year was still under two years old when it was used.

Table 4. Revisions to state growth by fiscal year

Fiscal year	State-years	States	Mean revision (pp)	Mean absolute revision (pp)	Revised down
2017-18	15	15	−0.97	1.73	73%
2018-19	14	14	−1.18	2.28	79%
2019-20	15	15	−2.25	2.30	93%
2020-21 (COVID)	14	14	+0.32	2.66	50%
2021-22 (COVID)	15	15	−0.31	2.14	60%
2022-23	16	16	−0.08	1.97	56%
2023-24	15	15	+0.93	1.36	33%
2024-25	17	17	+0.11	0.39	53%
2025-26	9	9	−0.12	0.13	56%
Headline years (2017-18 to 2019-20, 2022-23)	60	18	−1.10	2.07	

Notes: Revision = latest growth minus the state's own first estimate published within 12 months of the year's end. Recent years have had less time to be revised. Source: ITIHAS; authors' calculations.

6. Nowcast results

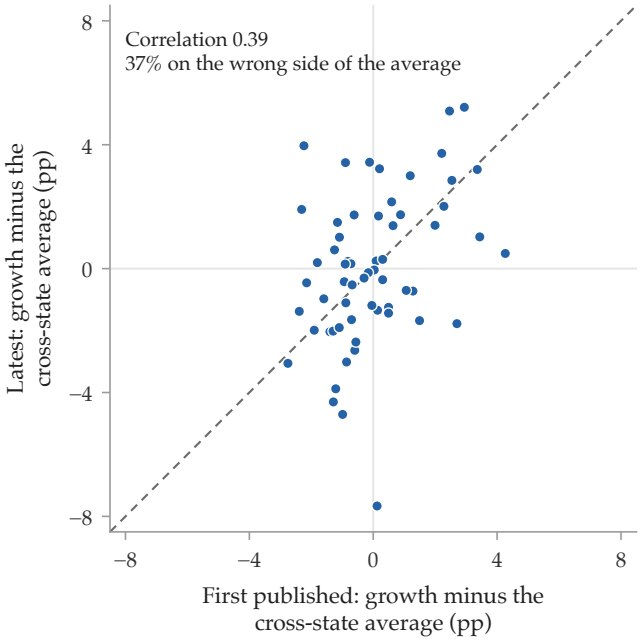
None of the 19 models was statistically better than the national-rate guess after the correction. The best attempts had about −4.6% error as compared to the benchmark, while e-way bills were significantly worse at +25.8% (electricity, at +8.2%, was not significant after the correction). For the best models, the 95% intervals ruled out any model being better by more than about 11–18% in squared error (Table 5). Figure 5 shows the same results for every model. None of the intervals lies fully below zero, which would mean a model is clearly better than the benchmark, while four lie fully above it. In the blind test of 2016-17 and 2017-18, no model beat the benchmark. Leaving out 2024-25, whose figures have not yet settled, no model wins either (Tables A2 and A3; the second is a robustness check added after the main results). Therefore, none of the 19 models built on public data could fix early figures, so the fix lies in the rules.

7. The Money

7.1 The 15th Commission

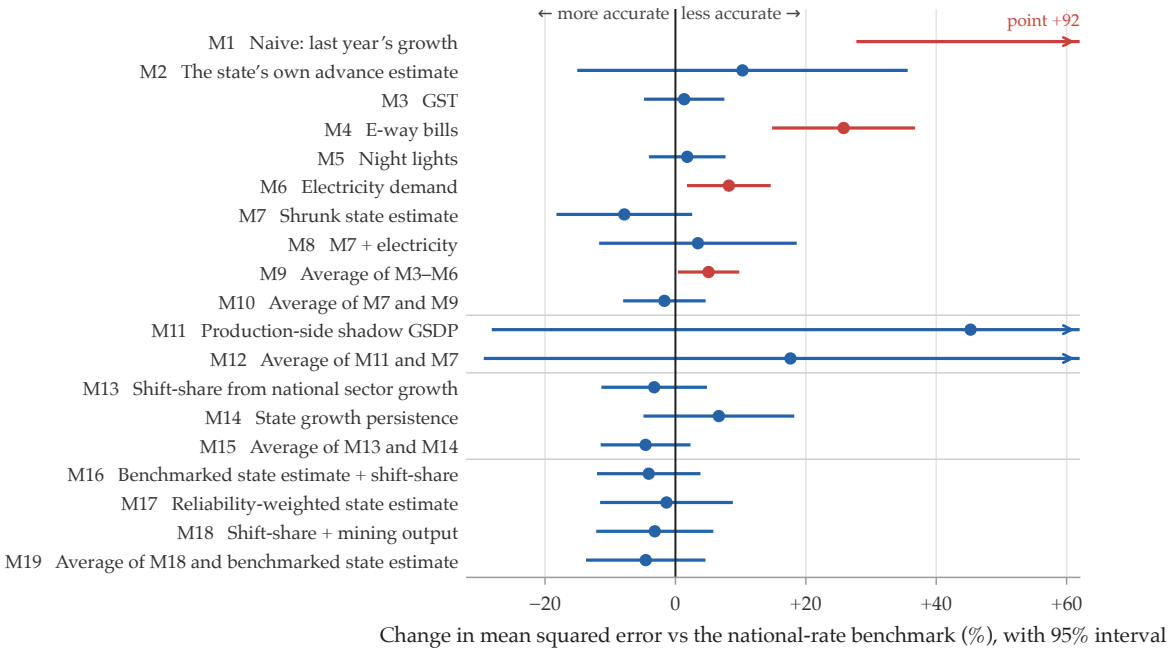
We took the Commission's own formula and swapped in today's GDP figures for the three years it used, and re-ran it and compared each state's share. Using the corrected figures, ₹25,654 crore would have gone to different states over 2021-26, which is about ₹5,131 crore

Figure 4. First-published versus latest relative growth



Notes: Each dot is a state-year: 60 state-years, 18 states, FY2017-18 to 2019-20 and 2022-23; each state's own first release within 12 months of the year's end. Dashed line: first print equals latest. Source: ITIHAS; authors' calculations.

Figure 5. Nowcast models against the national-rate benchmark



Notes: Each row is one of the 19 pre-specified models. The dot is the change in mean squared error compared with the benchmark, where every state grows at the national First Advance Estimate, and the line is the 95% interval, clustered by state. Red means the interval lies fully above zero. Arrows mean the interval continues past the edge. No model is better after the Holm correction. Evaluation years are FY2018-19, 2019-20, 2022-23, 2023-24 and 2024-25, and the horizontal lines separate the four rounds. Source: ITIHAS; authors' calculations.

Table 5. Nowcast models and their errors

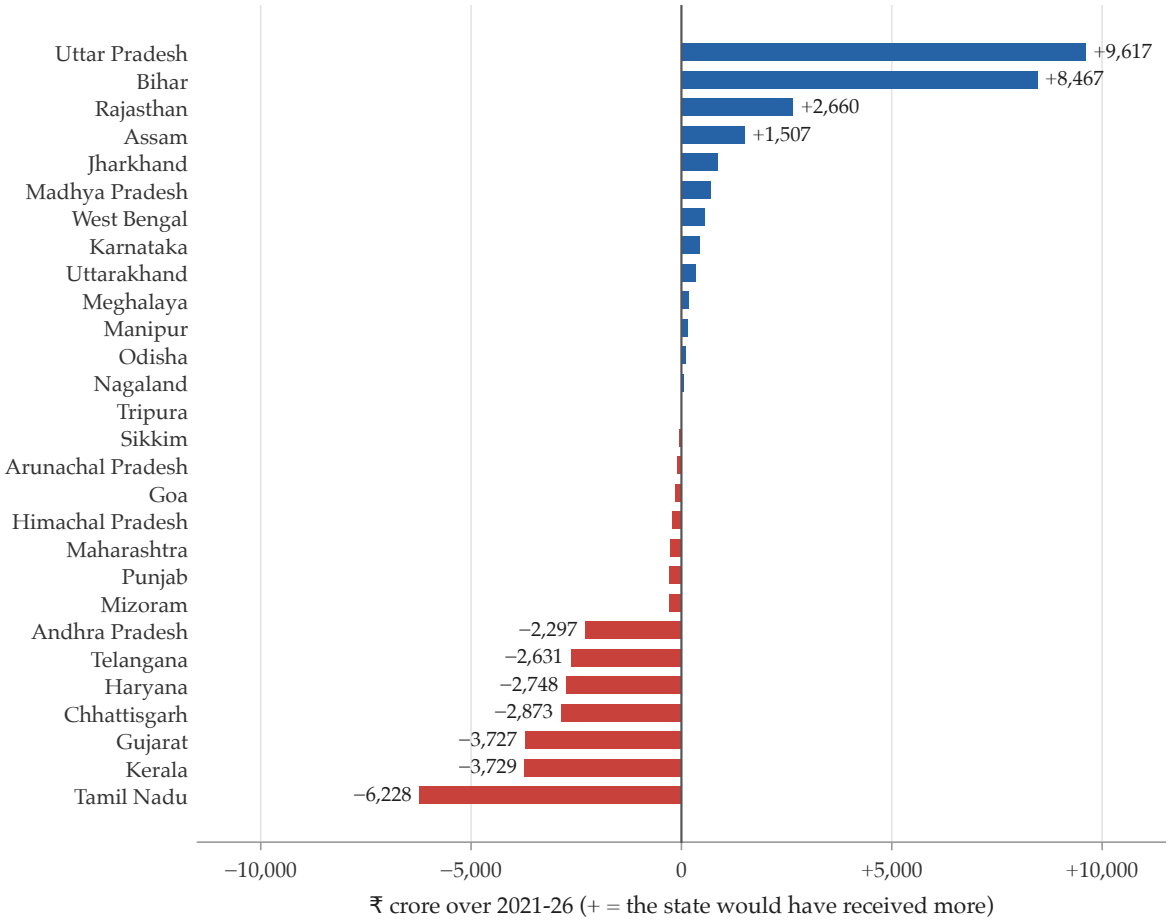
Model	Description	Round	n	RMSE (pp)	RMSE, national rate (pp)	MSE change (%)	95% CI (%)	Holm p
M1	Naive: last year's growth	1	88	3.25	2.35	+92.2	[+27.8, +156.7]	0.09
M2	The state's own advance estimate	1	47	2.50	2.38	+10.3	[−15.1, +35.6]	1.00
M3	GST	1	60	2.27	2.26	+1.3	[−4.8, +7.5]	1.00
M4	E-way bills	1	40	2.11	1.88	+25.8	[+14.8, +36.8]	<0.01
M5	Night lights	1	60	2.28	2.26	+1.8	[−4.1, +7.7]	1.00
M6	Electricity demand	1	58	2.39	2.30	+8.2	[+1.8, +14.6]	0.21
M7	Shrunk state estimate	1	31	2.38	2.48	−7.8	[−18.2, +2.6]	1.00
M8	M7 + electricity	1	31	2.52	2.48	+3.4	[−11.7, +18.6]	1.00
M9	Average of M3–M6	1	60	2.32	2.26	+5.1	[+0.4, +9.8]	0.55
M10	Average of M7 and M9	1	31	2.46	2.48	−1.7	[−8.0, +4.6]	1.00
M11	Production-side shadow GSDP	2	60	2.72	2.26	+45.3	[−28.2, +118.7]	1.00
M12	Average of M11 and M7	2	31	2.69	2.48	+17.6	[−29.4, +64.7]	1.00
M13	Shift-share from national sector growth	3	99	2.24	2.28	−3.3	[−11.4, +4.8]	1.00
M14	State growth persistence	3	99	2.35	2.28	+6.7	[−4.9, +18.2]	1.00
M15	Average of M13 and M14	3	99	2.22	2.28	−4.6	[−11.5, +2.3]	1.00
M16	Benchmarked state estimate + shift-share	4	100	2.22	2.27	−4.1	[−12.0, +3.8]	1.00
M17	Reliability-weighted state estimate	4	100	2.25	2.27	−1.4	[−11.6, +8.8]	1.00
M18	Shift-share + mining output	4	100	2.23	2.27	−3.2	[−12.1, +5.8]	1.00
M19	Average of M18 and benchmarked state estimate	4	100	2.21	2.27	−4.6	[−13.7, +4.6]	1.00

Notes: Benchmark (M0): every state grows at the national First Advance Estimate. Negative MSE change = more accurate than the benchmark. CIs clustered by state. Holm correction across all 19 comparisons. A model wins only if Holm $p < 0.05$ and the year-clustered estimate has the same sign: none does. Evaluation years FY2018-19, 2019-20, 2022-23, 2023-24, 2024-25; real-time training targets. Source: ITIHAS; authors' calculations. Years covered by each model are in Table A1.

each year (Figure 6). As a check, we also measured the revisions from the published edition whose figures are closest to the Commission's own, the November 2021 Handbook. On that base, ₹10,154 crore would have moved over 2021-26, about ₹2,031 crore a year (Table 6, Panel C). We keep ₹25,654 crore as the main figure because the base edition was fixed before any results were seen. If corrected, the gains would mostly go to Uttar Pradesh and Bihar, who would receive about 70% of the moved money. Losses would be spread across states near the top.

A state's money can move even if its GDP does not because income distance is measured against a benchmark state (equations 4 and 5). For the 15th Commission, the measuring stick was Haryana. Haryana's 2018-19 GDP was later cut by 4.8%, shortening the stick. A state just below it, such as Tamil Nadu, lost a large part of its distance, whereas a state far below it like Bihar barely noticed it. Therefore, money shifts from states near the top to the poorest ones, even if their own numbers never change. Tamil Nadu's GDP was not revised at all, yet it would have received ₹6,228 crore less, and Bihar's 2018-19 GDP was revised by

Figure 6. Money that would have gone to different states under the 15th Commission, by state



Notes: 15th Finance Commission, 2021-26 award. Each GSDP figure the Commission used (2016-17 to 2018-19) is scaled by its later revision (RBI Handbook Oct 2020 to the latest MoSPI/RBI figure); all other inputs and the formula are unchanged. ₹ at actual devolution, 2021-26. Source: ITIHAS; authors' calculations.

only −0.45%, yet it would have received ₹8,467 crore more. A state's money depends on other states' statistics, which it cannot control. About 78% of the money comes from 2018-19, which is the youngest year the Commission used, which links back to the two-year rule (Table 6).

7.2 *Is it normal?*

Verification is required as a single Commission could be a fluke. Placebo Commissions are the same formula with revisions from each older RBI edition that have the same data ages. We found a range of ₹2,787–6,781 crore a year, with the real ₹5,131 crore close to the typical ₹5,044 crore, showing us that it is normal (Figure 7, Table 7). There were 5,000 random mixes with the real case at the 22nd percentile, so it was, if anything, mild. No state can plan for it as the range includes both gaining and losing (Table 8).

7.3 *What's coming*

The GDP years of the 16th Commission are already being corrected, and ₹1,110 crore a year has moved after only 8 months (MoSPI's August 2025 figures compared with April 2026)

Table 6. Money that would have gone to different states, by Finance Commission award

Award	Data used (V)	Cells revised	Mean absolute level revision (%)	Max (%)	Share moved (pp)	₹ crore, whole award	₹ crore a year
Panel A. All data years revised							
15th FC interim (2020-21)	Handbook Mar 2019	67	1.53	16.7	0.641	3,812	3,812
15th FC final (2021-26)	Handbook Oct 2020	73	1.24	12.3	0.455	25,654	5,131
16th FC (2026-31)	MoSPI Aug 2025	62	0.33	9.1	0.073	5,549	1,110
Panel B. One data year revised at a time							
15th FC final (2021-26)	2016-17				0.046	2,611	
15th FC final (2021-26)	2017-18				0.098	5,546	
15th FC final (2021-26)	2018-19				0.352	19,875	
15th FC interim (2020-21)	2015-16				0.205	1,219	
15th FC interim (2020-21)	2016-17				0.256	1,524	
15th FC interim (2020-21)	2017-18				0.302	1,798	
16th FC (2026-31)	2022-23				0.019	1,486	
16th FC (2026-31)	2023-24				0.053	4,080	
Panel C. Sensitivity: base = the closest-matching later edition							
15th FC interim (2020-21)	Handbook Oct 2020	65	0.33	4.8	0.160	950	950
15th FC final (2021-26)	Handbook Nov 2021	64	0.33	4.3	0.180	10,154	2,031

Notes: Share moved = half the sum of absolute changes in states' shares of the divisible pool. Panel B omits 16th FC data years that have not been revised (share moved 0). Latest data (L): MoSPI Apr 2026, else RBI Handbook Dec 2025. ₹ at actual devolution (2026-27 budget estimate for the 16th FC). Panel B ₹ figures are for the whole award (16th FC: 5 × the 2026-27 budget estimate). Source: ITIHAS; authors' calculations.

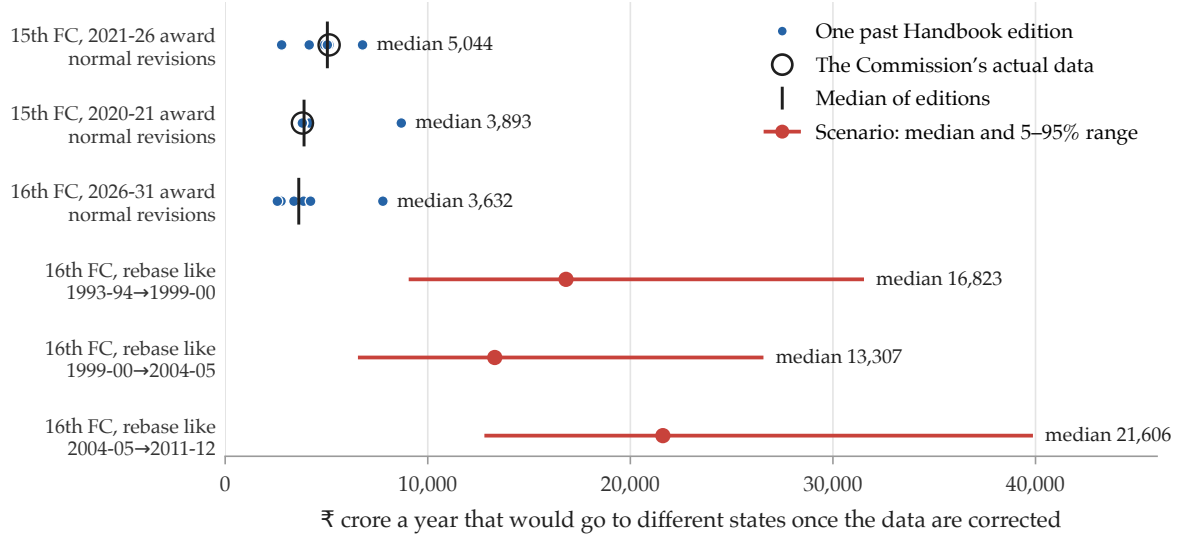
(Table 6). Most of this comes from the 2023-24 figures of three states: Goa cut 8.2%, Uttarakhand cut 5.4%, and Odisha raised 2.7%. Its newest years are under two years old, so the two-year rule tells us that more corrections are coming, with past Commissions suggesting ₹3,400–3,900 crore a year once they settle (the May 2018 and March 2019 Handbooks cover the same data years; Table 7). The 16th Commission gives 10% of the money by each state's share of India's GDP ("contribution to GDP"). For Karnataka, Telangana, Haryana, Goa and Sikkim, a higher GDP figure now brings more money rather than less (Table 8, last column). Under the new factor, bigger means more money; therefore, for these five states, the new factor wins. However, most states will still lose money when their GDP is raised. The larger risk is the new base year which was moved to 2022-23 in February 2026, and state figures on the new base, due by 30 September 2026, have not yet been published (Rajya Sabha, 2026).

Table 7. Placebo Commissions: the same formula with older RBI Handbook editions

Formula	Edition (placebo data)	Data years	Mean absolute level revision (%)	Share moved (pp)	₹ crore a year
15th FC final (2021-26)	Handbook Jun 2017	2013-14, 2014-15, 2015-16	2.32	0.435	4,907
15th FC final (2021-26)	Handbook May 2018	2014-15, 2015-16, 2016-17	2.08	0.601	6,781
15th FC final (2021-26)	Handbook Mar 2019	2014-15, 2015-16, 2016-17	0.85	0.452	5,100
15th FC final (2021-26)	Handbook Oct 2020	2016-17, 2017-18, 2018-19	1.24	0.455	5,131 (actual)
15th FC final (2021-26)	Handbook Nov 2021	2017-18, 2018-19, 2019-20	1.16	0.367	4,148
15th FC final (2021-26)	Handbook Nov 2022	2018-19, 2019-20, 2020-21	1.99	0.447	5,044
15th FC final (2021-26)	Handbook Nov 2023	2019-20, 2020-21, 2021-22	1.06	0.247	2,787
15th FC final (2021-26): summary	median of 7 editions				5,044
	state resampling, median [5–95%]				6,467 [3,774–9,394]
15th FC interim (2020-21)	Handbook Jun 2017	2014-15, 2015-16, 2016-17	2.59	0.668	3,974
15th FC interim (2020-21)	Handbook May 2018	2015-16, 2016-17, 2017-18	2.28	0.699	4,159
15th FC interim (2020-21)	Handbook Mar 2019	2015-16, 2016-17, 2017-18	1.53	0.641	3,812 (actual)
15th FC interim (2020-21)	Handbook Oct 2020	2017-18, 2018-19, 2019-20	2.07	1.461	8,694
15th FC interim (2020-21)	Handbook Nov 2021	2018-19, 2019-20, 2020-21	2.04	0.637	3,788
15th FC interim (2020-21)	Handbook Nov 2022	2019-20, 2020-21, 2021-22	2.80	0.641	3,811
15th FC interim (2020-21): summary	median of 6 editions				3,893
	state resampling, median [5–95%]				5,020 [3,413–10,601]
16th FC (2026-31)	Handbook May 2018	2011-12, 2012-13, 2014-15, 2015-16, 2016-17	1.28	0.510	7,783
16th FC (2026-31)	Handbook Mar 2019	2011-12, 2012-13, 2014-15, 2015-16, 2016-17	0.53	0.222	3,395
16th FC (2026-31)	Handbook Oct 2020	2013-14, 2014-15, 2016-17, 2017-18, 2018-19	0.77	0.254	3,870
16th FC (2026-31)	Handbook Nov 2021	2014-15, 2015-16, 2017-18, 2018-19, 2019-20	0.73	0.181	2,757
16th FC (2026-31)	Handbook Nov 2022	2015-16, 2016-17, 2018-19, 2019-20, 2020-21	1.24	0.276	4,215
16th FC (2026-31)	Handbook Nov 2023	2016-17, 2017-18, 2019-20, 2020-21, 2021-22	0.68	0.169	2,576
16th FC (2026-31): summary	median of 6 editions				3,632
	state resampling, median [5–95%]				3,882 [2,428–7,145]

Notes: State resampling: 5,000 draws; in each, every state takes its whole revision vector from one randomly chosen edition. The actual 15th FC final case is at the 22nd percentile of the resampling distribution; under the rule fixed in advance it is typical, not an outlier. Source: ITIHAS; authors' calculations.

Figure 7. Placebo Commissions and the base-year scenario



Notes: Blue dots: placebo Commissions, the formula re-run with the revisions that later hit one past RBI Handbook edition, at the same data ages as the Commission's own data. Red: rebase scenario, 2,000 draws in which past rebase shifts in states' relative GSDP are drawn among similar-sized states; median and 5–95% range. Source: ITIHAS; authors' calculations.

If states shift as much as in past base changes, ₹13,000–22,000 crore a year could move, which is four to six times the normal amount (red points in Figure 7). The above is a scenario, not a forecast.

7.4 Borrowing caps

Each state's yearly borrowing cap is a % of its projected GDP (Section 2.2). We work it out twice for 2019-20 to 2024-25: once with the GDP the ministry had when it set the cap, and once with today's figures. This can be studied through the following equation:

$$NBC_{st} = c_t \times G_{s,t-2}\,(1+\bar{g}_s)^2\,, \qquad \bar{g}_s = \frac{1}{3}\left(g_{s,t-2} + g_{s,t-3} + g_{s,t-4}\right) \tag{9}$$

NBC_{st} is the borrowing cap of states for year t and c_t is the cap rate, which is 3% in most years, 4% in 2021-22 and 3.5% in 2022-23 (we leave out 2020-21's extra 2% for COVID and the 0.5% power-sector allowance). $G_{s,t-2}$ is the GDP from two years earlier, which is the newest available figure when the cap is set, and $\bar{g}_s$ is the average growth over years t−4 to t−2.

The caps which were built on the early figures were ₹1.44 lakh crore higher in total than the corrected figures would give (Table 9). A typical state's cap is off by 4.6% in a year. This differs from the tax split because it is not mainly money moving between the states. Most caps were too high, so states together got more room to borrow. Most of it falls in 2021-22 and 2022-23. The reasons for this are that those caps were projected from 2019-20 and 2020-21 GDP, the first of which is the slowdown year whose early figures were too high (Section 1). Moreover, the cap rate was raised to 4% and 3.5% in those two years, so each rupee of wrong GDP counted for more. The largest single case is Maharashtra in the 2021-22 period which had ₹12,485 crore, which is about 9% of its cap. The ministry does not publish its inputs,

Table 8. Money moved by state, per year

State	15th FC, actual (₹ cr/yr)	15th FC, placebo median [5–95%]	15th FC, ₹ per +1% own GSDP	16th FC, placebo median [5–95%]	16th FC, ₹ per +1% own GSDP
Andhra Pradesh	−459	+134 [−546, +957]	−327	+195 [−320, +803]	−393
Arunachal Pradesh	−18	+1 [−28, +58]	−8	+3 [−13, +26]	−6
Assam	+301	+17 [−362, +449]	−114	+7 [−241, +249]	−121
Bihar	+1,693	+262 [−1,697, +3,036]	−169	+121 [−1,426, +1,476]	−170
Chhattisgarh	−575	−4 [−667, +1,243]	−115	−49 [−459, +821]	−121
Goa	−32	−12 [−71, +30]	0	−5 [−24, +16]	+10
Gujarat	−745	−526 [−1,954, +677]	−477	+30 [−446, +553]	−534
Haryana	−550	−188 [−1,214, +506]	+186	−64 [−423, +314]	+82
Himachal Pradesh	−46	+8 [−153, +148]	−60	+28 [−47, +99]	−53
Jharkhand	+173	+213 [−444, +1,015]	−105	+41 [−332, +531]	−107
Karnataka	+86	−184 [−3,194, +1,723]	−545	−121 [−976, +836]	+331
Kerala	−746	+37 [−1,054, +960]	−300	+58 [−510, +642]	−328
Madhya Pradesh	+142	+76 [−844, +1,093]	−263	−1 [−569, +616]	−303
Maharashtra	−56	+910 [−2,036, +3,637]	−943	+1,320 [−647, +2,841]	−1,103
Manipur	+29	+17 [−31, +86]	−10	+7 [−21, +35]	−6
Meghalaya	+36	+12 [−33, +64]	−11	+3 [−19, +25]	−8
Mizoram	−61	−9 [−167, +15]	−8	−1 [−56, +8]	−4
Nagaland	+11	0 [−11, +14]	−10	+1 [−6, +8]	−6
Odisha	+21	−100 [−501, +480]	−184	−129 [−334, +143]	−207
Punjab	−57	−17 [−391, +325]	−191	−2 [−162, +159]	−195
Rajasthan	+532	+46 [−397, +593]	−318	+62 [−210, +415]	−361
Sikkim	−14	−5 [−30, +12]	0	+4 [−6, +28]	+7
Tamil Nadu	−1,246	−60 [−1,814, +1,444]	−612	−21 [−961, +993]	−723
Telangana	−526	−14 [−1,022, +506]	−305	−73 [−517, +422]	+129
Tripura	+4	−3 [−80, +29]	−18	−1 [−42, +16]	−14
Uttar Pradesh	+1,923	−1,648 [−4,680, +2,046]	−453	−1,887 [−4,057, +132]	−473
Uttarakhand	+68	+92 [−228, +375]	−87	+90 [−74, +228]	−80
West Bengal	+110	+256 [−606, +2,356]	−372	+198 [−345, +1,686]	−436

Notes: ₹ crore a year; + = the state would receive more. Actual: the 15th FC 2021-26 counterfactual (Figure 6) divided by 5. Placebo ranges: state resampling of placebo Commissions (5,000 draws). ₹ per +1%: the change in the state's own devolution if its GSDP were 1% higher in every data year; valid for small changes only (the formula has rank kinks). Source: ITIHAS; authors' calculations.

and our version of the rule matches its caps within a median of about 2% in 2024-25 but is a median of 15.5% off in 2021-22.

8. The fix

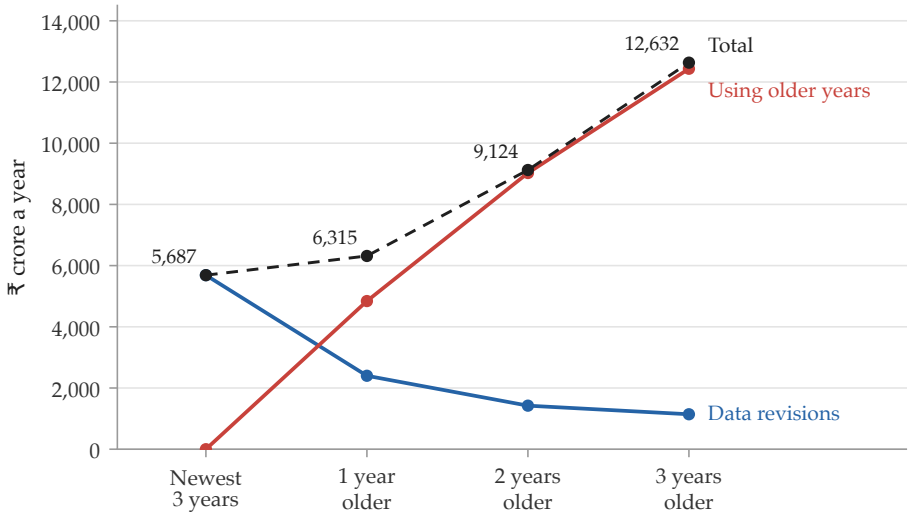
The obvious fix is to use older years that have settled, and we tested it with a rule set in advance. One year older gave us ₹6,315 crore a year of error against ₹5,687 for the Commission's current window, and was better in only 1 of 5 past cases (Figure 8). The ₹5,687 crore here differs from the ₹5,131 crore in Table 6 because Figure 8 changes only the income-distance criterion, measures the error against the Commission's window in mature data, and averages over five Handbook editions. This was because older data removes some correction error but adds staleness, given that relative incomes move about as fast as the figures get corrected.

Table 9. Borrowing ceilings under early and corrected GSDP

Fiscal year	States	Normal ceiling (% of GSDP)	Gross |Δ ceiling| (₹ crore)	Net Δ (₹ crore)	Gross, % of all ceilings	Median |Δ| per state (%)	Gap vs ministry (median %)
2019-20	33	3%	34,825	+1,785	5.4	2.9	–
2020-21	33	3%	31,068	−19,040	4.3	4.4	7.5
2021-22	33	4%	78,330	−69,954	7.5	6.5	15.5
2022-23	33	3.5%	55,785	−42,088	6.7	5.0	5.5
2023-24	33	3%	43,968	−949	5.2	4.4	7.2
2024-25	33	3%	37,063	−13,889	3.6	3.5	2.0
2019-20 to 2024-25			281,040	−144,135	5.5	4.6	

Notes: Ceiling = c × projected GSDP, projected as $G(t-2) \times (1 + \bar{g})^2$ (14th FC, paras 14.65–14.66). Δ = ceiling from the latest GSDP minus ceiling from the earliest central GSDP containing G(t−2). Net negative: ceilings built on first-published GSDP were higher. Last column: how closely the rule applied to the earliest data reproduces the GSDP implied by the ministry's state-wise ceilings (Lok Sabha USQ 2380, 4 Aug 2025); the ministry's own inputs are not public, so read this table as the rule's sensitivity. Covers 28 states, Delhi, Puducherry, Chandigarh, the Andaman and Nicobar Islands, and Jammu and Kashmir. Source: ITIHAS; authors' calculations.

Figure 8. Error from using older data years



Notes: 15th FC income-distance criterion, other criteria fixed. Target: the Commission's own 3-year window measured with mature data. Mean over five RBI Handbook editions (Jun 2017 to Nov 2021; four for the oldest point). ₹ at the 2021-26 average pool. Source: ITIHAS; authors' calculations.

Partway through the five years, we re-ran the same formula with the corrected figures for the same years, using the new shares for the remaining years, without any states paying anything back. We found that if done in year 3, it fixes a median 44% of the money that would otherwise move (range 31–60%) (Table 10). Year 2 is too early where the figures have not yet settled, so it fixes only 35%, and year 4 is too late as few years are left to apply the new shares, fixing only 30%. For the 16th Commission, year 3 would be 2028-29.

The same idea applies for borrowing, where we recalculate the caps once the figures for those years are corrected. For the base-year change, before the new-base GDP is used for any money decision, a table linking old and new figures should be published and the shares both

Table 10. Revision error, staleness and the true-up

Quantity	Cells / n	Cross-state SD (%)
Panel A. Revisions and changes in relative income		
Revision to relative income, data <1 year old	71	4.71
Revision to relative income, data 1–2 years old	179	4.87
Revision to relative income, data 2–3 years old	222	2.93
Revision to relative income, data 3+ years old	1,373	0.60
Change in relative income over 1 year	10 year pairs	4.27
Change in relative income over 2 years	9 year pairs	6.35
Change in relative income over 3 years	8 year pairs	7.53
Panel B. Model: error of a 3-year window (SD, %). Middle column: revisions independent; right column: revisions fully correlated		
Newest 3 years (the Commission's design)	1.91	2.80
Lagged one year	2.71	2.86
Panel C. True-up in award year τ: share of the money corrected. Middle column: editions; right column: median [min–max]		
$\tau = 2$	7	35% [18–58%]
$\tau = 3$	7	44% [31–60%]
$\tau = 4$	6	30% [28–40%]

Notes: Panel A: relative income = log GSDP per person minus the cross-state mean; 28 states; mature data (FY $\leq$ 2021-22). Panel B: lagging helps only if the revision error it removes exceeds the staleness it adds; the model ranks the Commission's design first under both assumptions, as the money test (Figure 8) does. Panel C: shares re-fixed in award year τ with the Handbook edition about $\tau - 1$ years after each placebo edition; equal money each year. Source: ITIHAS; authors' calculations.

ways should be worked out so states can see what changed because of the new base rather than the economy.

9. Limitations

"Latest" is not final and recent years will keep changing with the new base year revising everything. Therefore, the 16th Commission number is early. The Commissions used MoSPI's special "comparable" GSDP, which matches no published edition exactly (the mean gap is 1.6–2.1%), so the amount that moves depends on which edition is taken as the starting point (Table 6, Panel C). West Bengal and Madhya Pradesh have no releases of their own, but rather only the central version. 249 values were read from scans and we hand-checked only a sample of them. The study only changes GDP and keeps everything in the formula fixed, which is a "what if" and not a claim about what the Commission would have done in practice. The borrowing rule is only roughly reproduced and for the five union territories it is applied mechanically, as the ministry's state-wise ceilings cover only the 28 states. The base-year number is a scenario, and the nowcast test covers only a few years. Better data in the future could change these results.

10. Conclusion

The study asked how much money has moved between states because the tax split and borrowing limits use GDP figures that have not settled, and what change to the rules would reduce the movement. We found that figures required about two years to settle; about ₹3,600–5,000 crore a year moves under each Commission; and borrowing caps were ₹1.44 lakh crore higher than corrected figures give, under our version of the rule. The fix is not that older data can help, but rather a year-3 true-up model, which reduces money movement by a median 44%. For policymakers, a true-up for the 16th Commission can be enacted in 2028-29, and a linking table should be published before the new base year is used. The database, ITIHAS, is free, and will be updated as new figures come out.

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Conflicts of interest

The authors declare no competing interests.

References

Diebold, F.X. and Mariano, R.S. (1995) 'Comparing predictive accuracy', *Journal of Business & Economic Statistics*, 13(3), pp. 253–263. doi:10.1080/07350015.1995.10524599.

Directorate of Economics and Statistics, Telangana (2019) *Gross State Domestic Product of Telangana State: Advance Estimates, 2018-19* (22 February 2019). Hyderabad: Government of Telangana. Available at: https://des.telangana.gov.in/publications/GSDP_Advance_Estimates_2018.pdf.

Fifteenth Finance Commission (2020) *Finance Commission in COVID Times: Report for 2021–26. Volume I: Main Report* (October 2020). New Delhi: Fifteenth Finance Commission, Government of India. Available at: https://fincomindia.nic.in/asset/doc/commission-reports/XVFC%20VOL%20I%20Main%20Report.pdf.

Fourteenth Finance Commission (2015) *Report of the Fourteenth Finance Commission*. New Delhi: Fourteenth Finance Commission, Government of India. Available at: https://fincomindia.nic.in/asset/doc/commission-reports/14th-FC/14thFCReport.pdf.

Frankel, J. (2011) 'Over-optimism in forecasts by official budget agencies and its implications', *Oxford Review of Economic Policy*, 27(4), pp. 536–562. doi:10.1093/oxrep/grr025.

Frankel, J. and Schreger, J. (2013) 'Over-optimistic official forecasts and fiscal rules in the eurozone', *Review of World Economics*, 149(2), pp. 247–272. doi:10.1007/s10290-013-0150-9.

Ghosh, B., Singh, N. and Kuruganti, S. (2026) 'Nowcasting state-level economic activity in India with night-time light and Google Trends', *Reserve Bank of India Bulletin*, October, pp. 183–198. Available at: https://www.rbi.org.in/Scripts/BS_ViewBulletin.aspx?Id=24519.

Holm, S. (1979) 'A simple sequentially rejective multiple test procedure', *Scandinavian Journal of Statistics*, 6(2), pp. 65–70. Available at: https://www.jstor.org/stable/4615733.

ICRA (2022) *State Government Finances: Downward Revision in States' GSVA/GSDP Estimates Could Lead to Inadvertent Fiscal Breaches, Highlighting Need to Improve GSVA/GSDP Estimation*. ICRA research report, November. Available at: https://icra.in/Rating/DownloadResearchSpecialCommentReport?id=4717.

Lok Sabha (2025) *Unstarred Question No. 2380: Net Borrowing Ceiling*, answered by the Minister of State in the Ministry of Finance (Department of Expenditure) on 4 August 2025, Annexure-I. New Delhi: Lok Sabha. Available at: https://sansad.in/getFile/lsapps/loksabhaquestions/annex/185/AU2380_3uo04y.pdf?source=lsapps.

Mankiw, N.G. and Shapiro, M.D. (1986) 'News or noise? An analysis of GNP revisions', *Survey of Current Business*, 66(5), pp. 20–25.

Ministry of Finance (2026a) *Budget at a Glance 2026–2027*. New Delhi: Ministry of Finance, Government of India, February. Available at: https://www.indiabudget.gov.in/doc/Budget_at_Glance/budget_at_a_glance.pdf.

Ministry of Finance (2026b) *Explanatory Memorandum as to the Action Taken on the Recommendations Made by the Sixteenth Finance Commission*. New Delhi: Budget Division, Ministry of Finance, Government of India, February. Available at: https://fincomindia.nic.in/asset/doc/commission-reports/16th-FC/16fc-EM.pdf.

Ministry of Statistics and Programme Implementation (2025) *Gross State Domestic Product at Current and Constant (2011-12) Prices, Base Year 2011-12: State-wise Estimates as on 1 August 2025* [data tables SDPAFY25141ANN and SDPAFY25142ANN]. New Delhi: National Statistics Office, MoSPI, released 8 August 2025. Available at: https://esankhyiki.mospi.gov.in.

Ministry of Statistics and Programme Implementation (2026a) *Press Note on New Series of GDP Estimates with Base Year 2022-23: Second Advance Estimates of Annual Gross Domestic Product for 2025-26 and Quarterly Estimates of Gross Domestic Product from Q1 (April–June) of 2022-23 to Q3 (October–December) of 2025-26*. New Delhi: National Accounts Division, National Statistics Office, 27 February. Available at: https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/feb/doc2026227806401.pdf.

Ministry of Statistics and Programme Implementation (2026b) *Gross State Domestic Product at Current and Constant (2011-12) Prices, Base Year 2011-12: State-wise Estimates as on 15 April 2026* [data tables SDPAFY26001ANN and SDPAFY26002ANN]. New Delhi: National Statistics Office, MoSPI, released 17 April 2026. Available at: https://esankhyiki.mospi.gov.in.

Ministry of Statistics and Programme Implementation (2026c) *Uniform Guideline for Compilation of Gross State Value Added (GSVA) Estimates with Base Year 2022-23*. New Delhi: National Statistics Office, 7 May. Press release available at: https://static.pib.gov.in/WriteReadData/specificdocs/documents/2026/may/doc202657865101.pdf; guideline at: https://www.mospi.gov.in/product/more/6-Documents.

Mitchell, J. and Shiroff, T. (2025) 'Are revisions to state-level GDP data in the US well behaved?', *Economics Letters*, 254, article 112407. doi:10.1016/j.econlet.2025.112407.

Narayana, D. (2024) *Horizontal Devolution: The Tyranny of Income Distance*. GIFT Discussion Paper 05/2024. Thiruvananthapuram: Gulati Institute of Finance and Taxation. Available at: https://gift.res.in/wp-content/uploads/2024/11/Horizontal-devolution-The-tyranny-of-Distance.pdf.

Rajya Sabha (2026) *Starred Question No. 234: New GDP Series and Methodology to Compute GSDP*, answered by the Ministry of Statistics and Programme Implementation on 10 August 2026. New Delhi: Rajya Sabha. Available at: https://sansad.in/getFile/annex/271/AS234_MGfYsu.pdf?source=pqars.

Reserve Bank of India (2017–2025) *Handbook of Statistics on Indian States*, editions 2016-17 to 2024-25 (released 24 June 2017; 5 May 2018; 3 March 2019; 13 October 2020; 24 November 2021; 19 November 2022; 15 November 2023; 9 December 2024; 11 December 2025). Mumbai: Reserve Bank of India. Available at: https://rbi.org.in/Scripts/AnnualPublications.aspx?head=Handbook%20of%20Statistics%20on%20Indian%20States.

Sixteenth Finance Commission (2025) *Report for 2026–31. Volume I: Main Report*. New Delhi: Sixteenth Finance Commission, Government of India. Available at: https://fincomindia.nic.in/asset/doc/commission-reports/16th-FC/reports/Vol1-Main-Report.pdf.

Appendix A. Additional results

Table A1. Evaluation years covered by each nowcast model (state-years)

Model	2018-19	2019-20	2022-23	2023-24	2024-25	Total
M1	16	13	19	20	20	88
M2	8	8	11	7	13	47
M3	–	–	20	20	20	60
M4	–	–	–	20	20	40
M5	–	–	20	20	20	60
M6	–	–	20	19	19	58
M7	–	–	11	7	13	31
M8	–	–	11	7	13	31
M9	–	–	20	20	20	60
M10	–	–	11	7	13	31
M11	–	–	20	20	20	60
M12	–	–	11	7	13	31
M13	19	20	20	20	20	99
M14	19	20	20	20	20	99
M15	19	20	20	20	20	99
M16	20	20	20	20	20	100
M17	20	20	20	20	20	100
M18	20	20	20	20	20	100
M19	20	20	20	20	20	100

Notes: Real-time training, as in Table 5. Models built on monthly indicators start in 2022-23 because national data vintages exist only from FY2018-19. Source: ITIHAS; authors' calculations.

Table A2. Blind test, FY2016-17 and 2017-18

Model	n	States	RMSE (pp)	RMSE, benchmark (pp)	MSE change (%)	95% CI (%)	Holm p
M2	17	11	2.10	2.38	−22.1	[−52.5, +8.4]	0.90
M13	40	20	2.92	2.84	+6.0	[−2.7, +14.8]	0.90
M16	40	20	2.87	2.84	+2.1	[−11.4, +15.5]	1.00
M17	40	20	2.87	2.84	+2.1	[−11.4, +15.5]	1.00
M18	40	20	2.96	2.84	+8.4	[−3.1, +19.9]	0.90
M19	40	20	2.90	2.84	+4.4	[−11.0, +19.8]	1.00

Notes: Only the six models of the final round were scored on these two years, fixed in advance. Holm correction across these six. No model wins. Source: ITIHAS; authors' calculations.

Table A3. Nowcast results with and without FY2024-25

Model	n	MSE change, all years (%)	n without 2024-25	MSE change without 2024-25 (%)	95% CI (%)	Holm p
M1	88	+92.2	68	+107.1	[+33.7, +180.4]	0.08
M2	47	+10.3	34	+26.7	[−0.7, +54.0]	1.00
M3	60	+1.3	40	−5.5	[−12.4, +1.4]	1.00
M4	40	+25.8	20	−16.0	[−32.8, +0.8]	1.00
M5	60	+1.8	40	−3.9	[−11.6, +3.8]	1.00
M6	58	+8.2	39	+3.0	[−5.1, +11.1]	1.00
M7	31	−7.8	18	−3.4	[−14.0, +7.2]	1.00
M8	31	+3.4	18	+13.1	[−3.9, +30.1]	1.00
M9	60	+5.1	40	−3.5	[−9.2, +2.2]	1.00
M10	31	−1.7	18	−2.9	[−10.9, +5.0]	1.00
M11	60	+45.3	40	+56.4	[−37.7, +150.4]	1.00
M12	31	+17.6	18	+24.4	[−36.6, +85.5]	1.00
M13	99	−3.3	79	−3.2	[−12.0, +5.7]	1.00
M14	99	+6.7	79	+4.6	[−8.9, +18.2]	1.00
M15	99	−4.6	79	−5.5	[−13.0, +2.0]	1.00
M16	100	−4.1	80	−3.0	[−11.3, +5.3]	1.00
M17	100	−1.4	80	+0.4	[−10.1, +10.8]	1.00
M18	100	−3.2	80	−3.4	[−13.1, +6.3]	1.00
M19	100	−4.6	80	−3.7	[−13.4, +6.1]	1.00

Notes: Robustness check added after the main results: the 2024-25 figures have not yet settled under the two-year rule. Real-time training; Holm correction across all 19 models. Without 2024-25, M4 rests on one year (2023-24). No model wins in either column. Source: ITIHAS; authors' calculations.

Table A4. State budgets' GSDP assumptions compared with the latest GSDP

Comparison	Years	n	States	Mean gap (%)	95% CI	Median gap (%)	Share above zero
Budget-year GSDP (budget estimate)	non-COVID years (headline)	90	20	+0.21	[−1.76, +2.19]	−0.05	0.49
Budget-year GSDP (budget estimate)	FY2017-18	10	10	+2.73		−1.03	0.40
Budget-year GSDP (budget estimate)	FY2018-19	1	1	−4.82		−4.82	0.00
Budget-year GSDP (budget estimate)	FY2019-20	6	6	+3.00		+3.43	0.50
Budget-year GSDP (budget estimate)	FY2022-23	20	20	−2.06		−1.68	0.35
Budget-year GSDP (budget estimate)	FY2023-24	20	20	−0.12		+0.70	0.50
Budget-year GSDP (budget estimate)	FY2024-25	20	20	−0.15		+0.67	0.55
Budget-year GSDP (budget estimate)	FY2025-26	13	13	+1.94		+1.10	0.69
Budget-year GSDP (budget estimate)	FY2020-21 (COVID, excluded)	17	17	+18.22		+19.82	1.00
Budget-year GSDP (budget estimate)	FY2021-22 (COVID, excluded)	15	15	+0.26		+0.23	0.53
Budget-year GSDP (budget estimate)	mature years only (FY≤ 2023-24)	57	20	−0.06		−0.60	0.42
Revised estimate for the year ending	non-COVID years (headline)	24	17	−0.17	[−2.22, +1.87]	+0.08	0.54
Revised estimate for the year ending	FY2023-24	16	16	+0.12		+0.32	0.56
Revised estimate for the year ending	FY2024-25	7	7	−1.29		0.00	0.43
Revised estimate for the year ending	FY2025-26	1	1	+2.94		+2.94	1.00

Notes: Gap = GSDP assumed in the state budget (current prices; PRS state budget analyses, CC BY 4.0) ÷ latest GSDP (MoSPI April 2026, else RBI Handbook December 2025) − 1. Confidence intervals clustered by state. COVID years shown but excluded from the headline rows. Source: PRS Legislative Research; ITIHAS; authors' calculations.